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JILIN UNIVERSITY



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Affective Vision Computing Lab

Demystifying Flux Architecture

Or Greenberg

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Paper Structure



◆ Main Content: 14 pages

- Abstract (0.5 page)
- Background (2.5 page)
- Method (8 page)
- Models and Tools (1 page)
- Reference (2 page)

◆ Supplementary: 4 pages



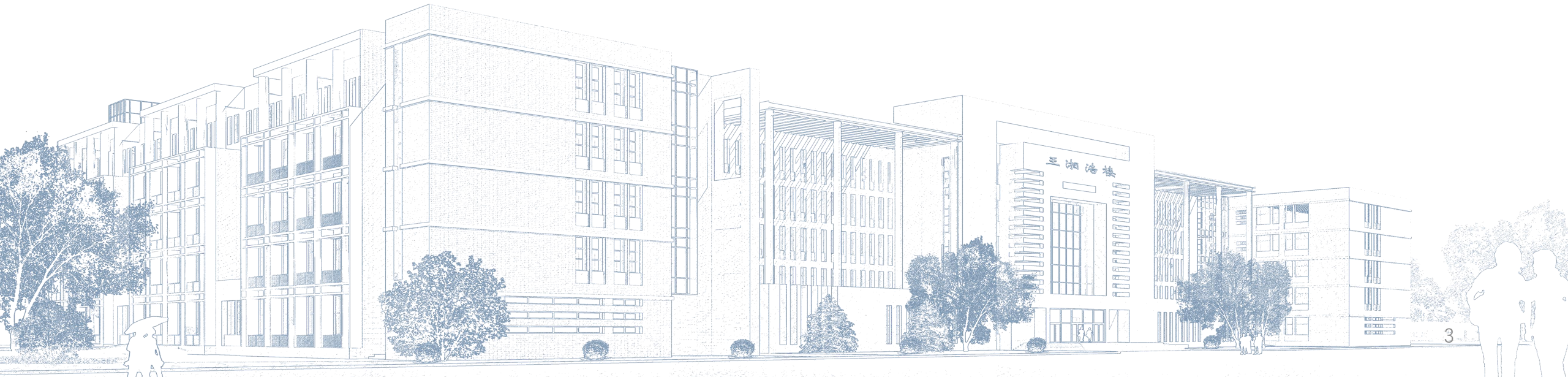
Outline

- ◆ Introduction & Motivation
- ◆ Background & Theoretical Foundations
- ◆ Method
- ◆ Qualitative Evaluation
- ◆ Conclusion



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01

Introduction & Motivation

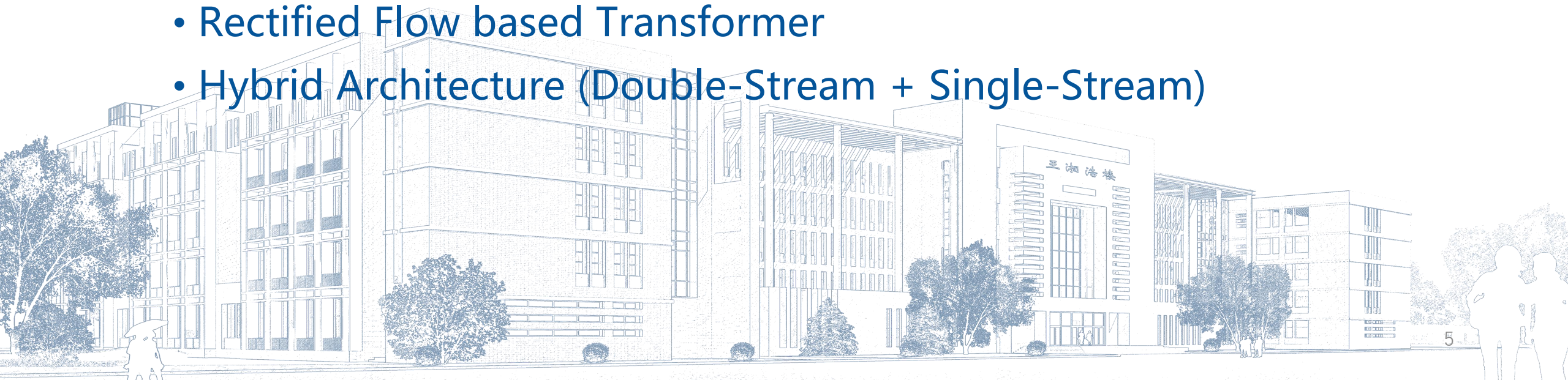


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Introduction & Motivation

- **Status of FLUX.1:** SOTA in text-to-image.
- **Motivation:** No official technical details released.
- **Contributions:** Demystifying FLUX via reverse engineering.
- **FLUX Structure:**
 - Rectified Flow based Transformer
 - Hybrid Architecture (Double-Stream + Single-Stream)

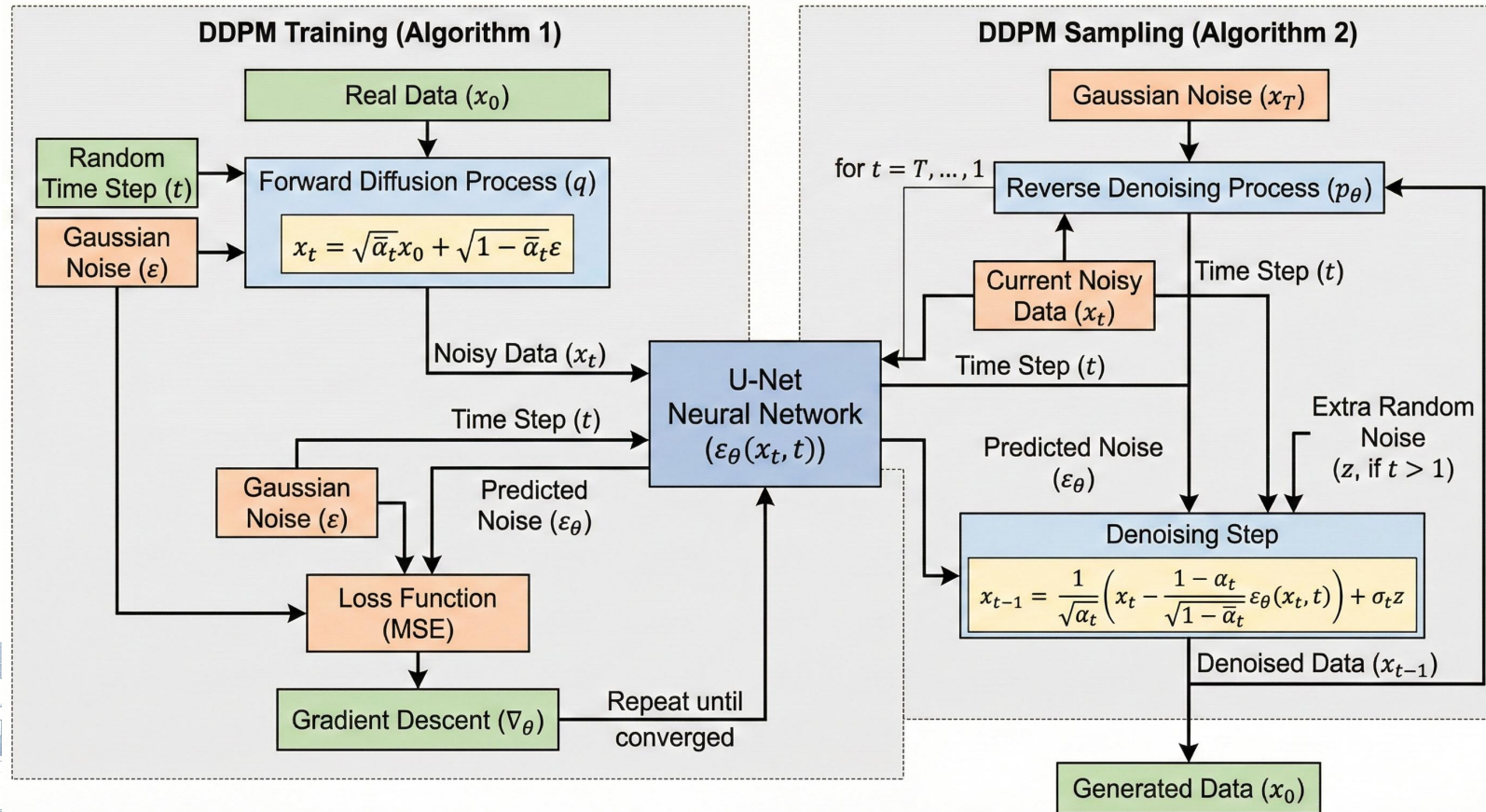


02

Background & Theoretical Foundations



➤ Diffusion Models(DDPM)



- Pixel Space
- U-Net

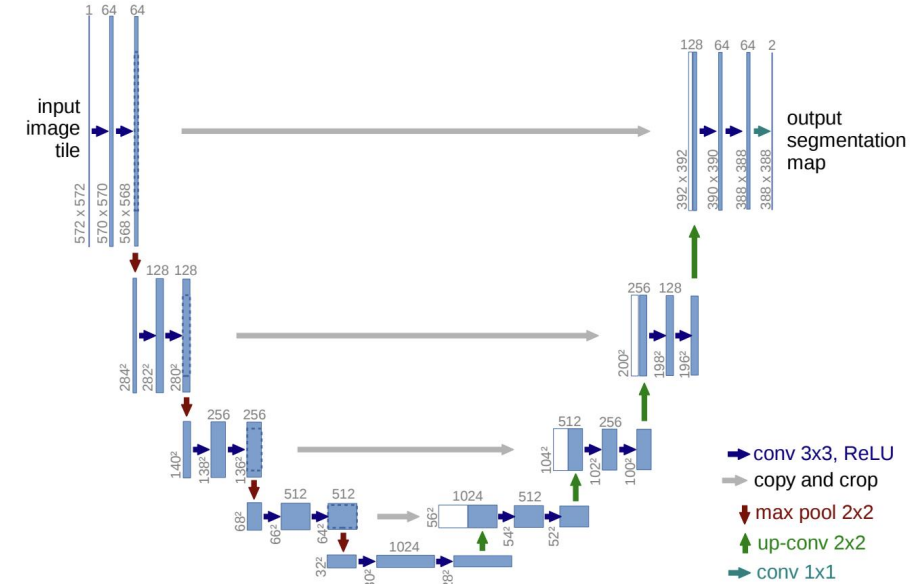
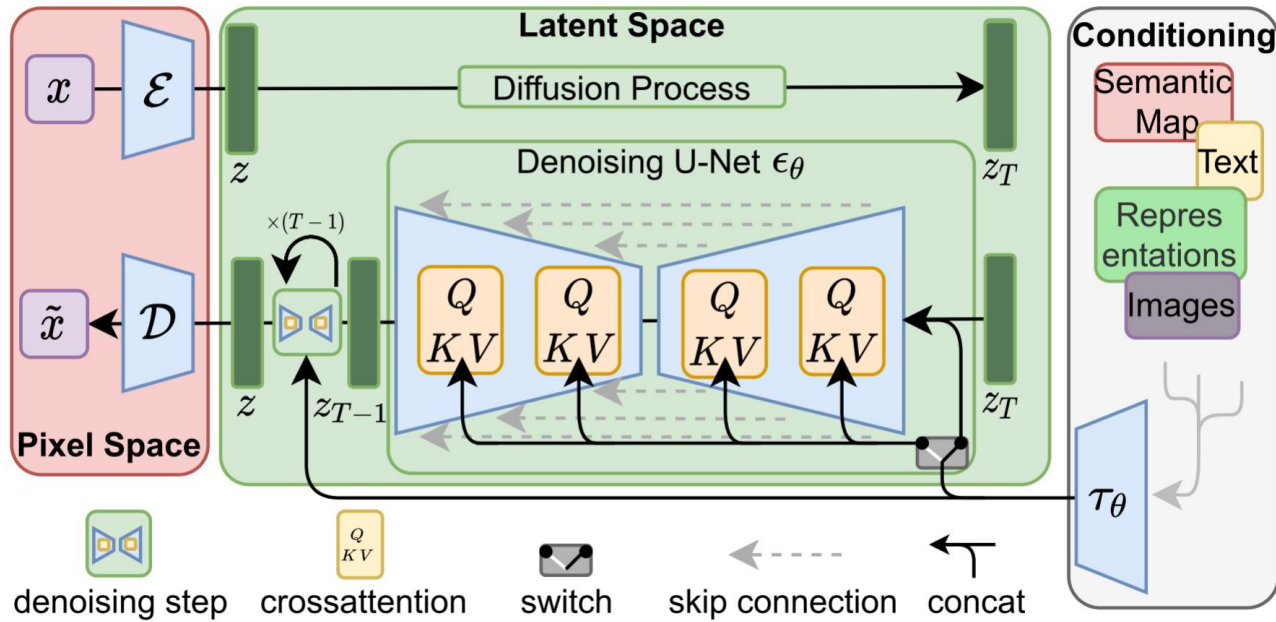
$$L_{\text{simple}}(\theta) := \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$



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➤ Latent Diffusion Models(LDM)

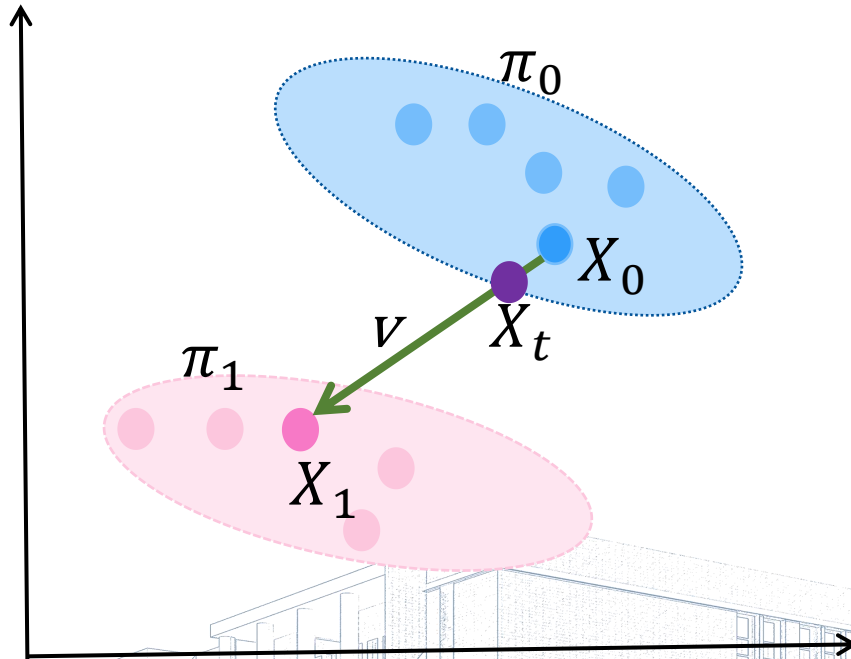


$$L_{LDM} := \mathbb{E}_{\mathcal{E}(x), \epsilon \sim \mathcal{N}(0,1), t} \left[\|\epsilon - \epsilon_\theta(z_t, t)\|_2^2 \right]$$

- Latent Space
- U-Net
- Cross-attention Layer



➤ Rectified Flows



Algorithm 1 Rectified Flow: Main Algorithm

Procedure: $Z = \text{RectFlow}((X_0, X_1))$:

Inputs: Draws from a coupling (X_0, X_1) of π_0 and π_1 ; velocity model $v_\theta: \mathbb{R}^d \rightarrow \mathbb{R}^d$ with parameter θ .

Training: $\hat{\theta} = \arg \min_{\theta} \mathbb{E} \left[\|X_1 - X_0 - v(tX_1 + (1-t)X_0, t)\|^2 \right]$, with $t \sim \text{Uniform}([0, 1])$.

Sampling: Draw (Z_0, Z_1) following $dZ_t = v_{\hat{\theta}}(Z_t, t)dt$ starting from $Z_0 \sim \pi_0$ (or backwardly $Z_1 \sim \pi_1$).

Return: $Z = \{Z_t : t \in [0, 1]\}$.

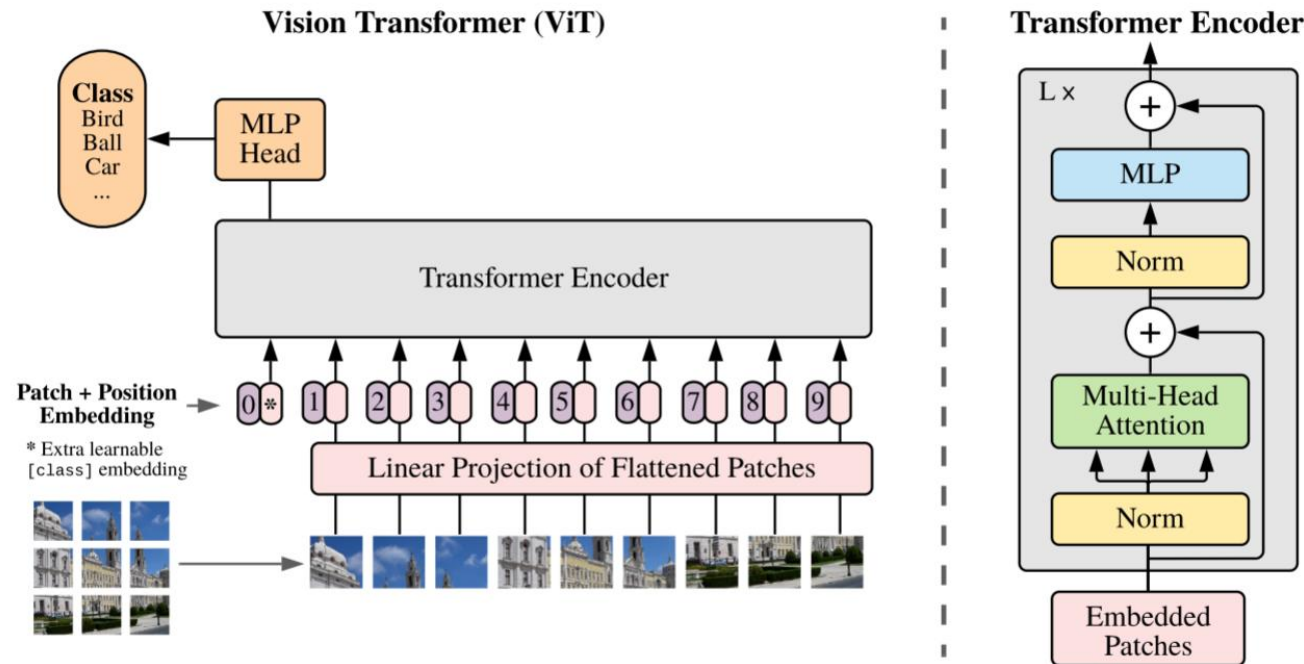
Reflow (optional): $Z^{k+1} = \text{RectFlow}((Z_0^k, Z_1^k))$, starting from $(Z_0^0, Z_1^0) = (X_0, X_1)$.

Distill (optional): Learn a neural network $\hat{T}$ to distill the k -rectified flow, such that $Z_1^k \approx \hat{T}(Z_0^k)$.

- Rectified Flow optimizes ODEs to follow straight trajectories between π_0 and π_1 .
- FLUX follows the Rectified Flow training paradigm.



➤ Vision Transformers



$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}} \cdot V)$$

- Convolution-Free: Replaces CNNs with a pure Transformer applied directly to image patch sequences.
- Sequence Modeling: Treats 2D image patches ($P \times P$) exactly like 1D tokens (words) in NLP.

03

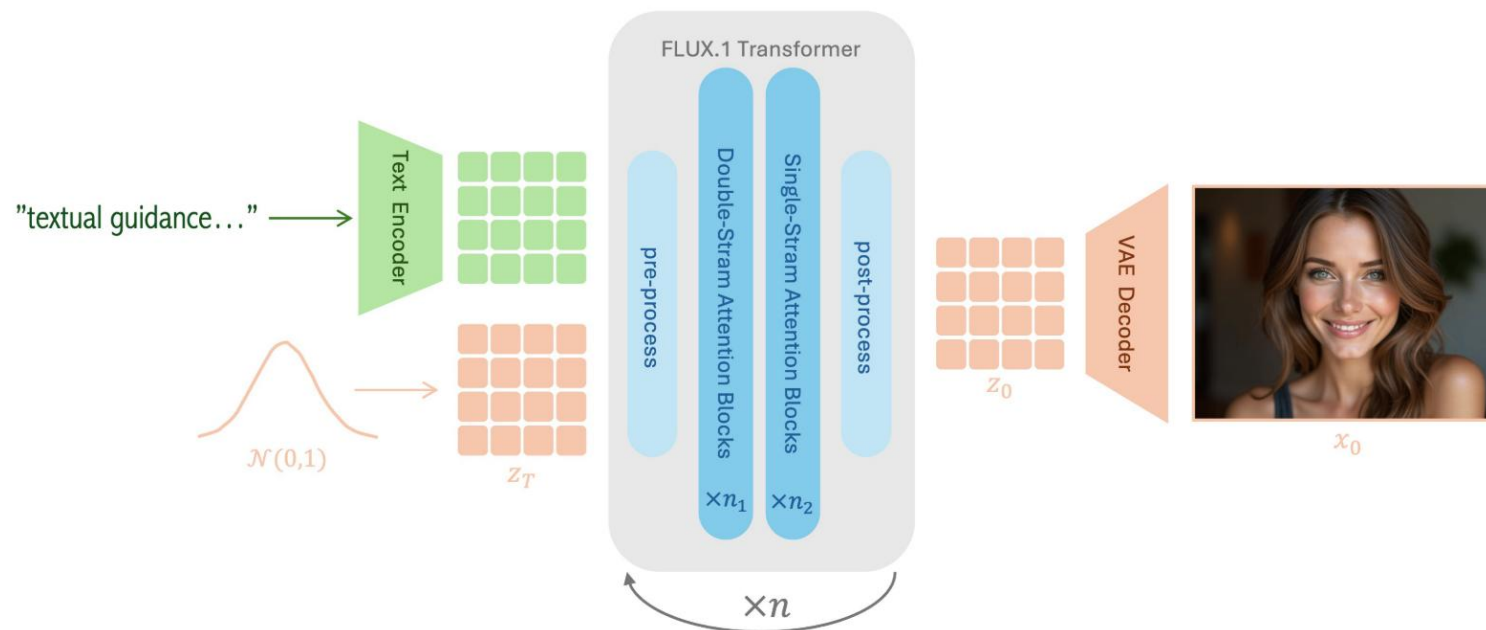
Method

➤ Overview



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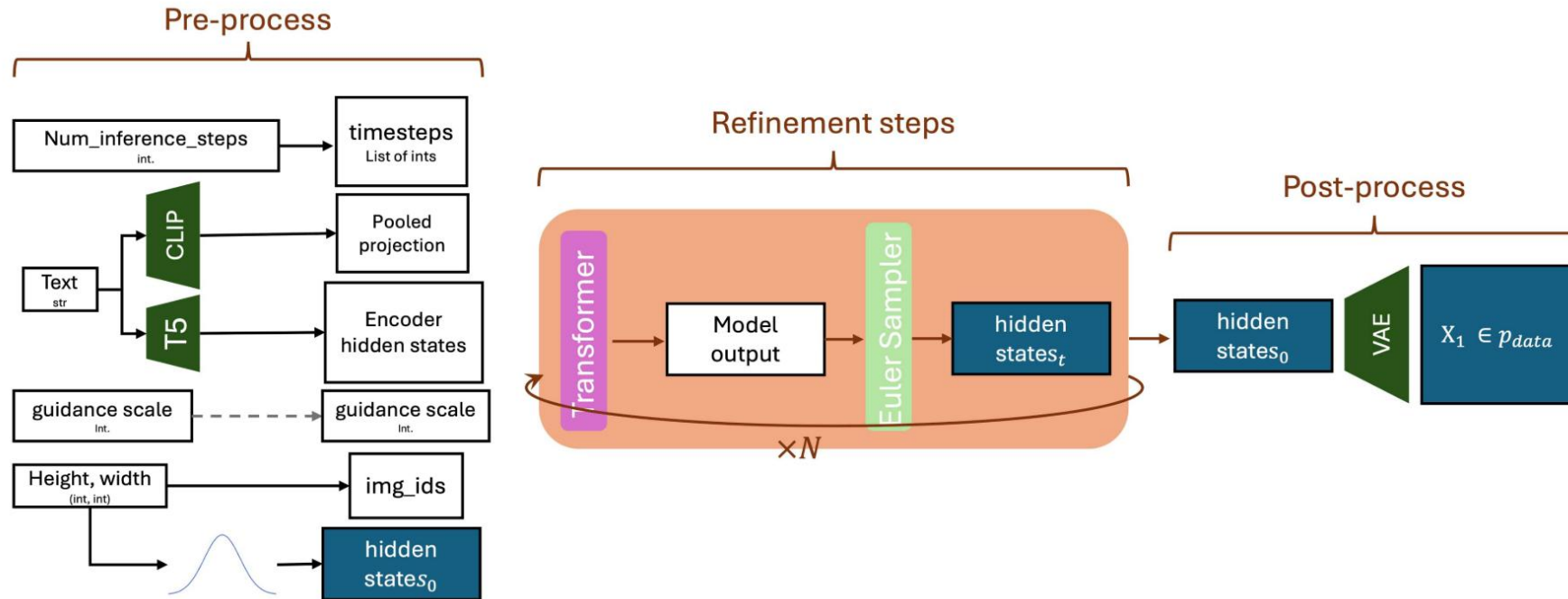
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- **Input:** Text embeddings+ Latent image embeddings
- **Output:** Text-conditioned image
- **Architectural Paradigm:** Transformer-based architecture (DiT).
- **Training Method:** Rectified Flow → Predict the velocity vector v



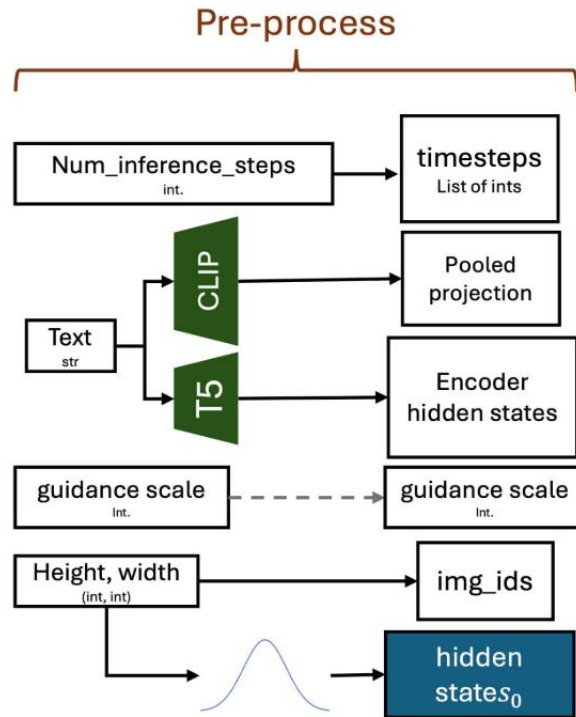
➤ FLUX.1 Sampling Pipeline



- ① Initiation and Pre-processing
- ② Iterative Refinement
- ③ Post-processing.



➤ Initiation and Pre-processing



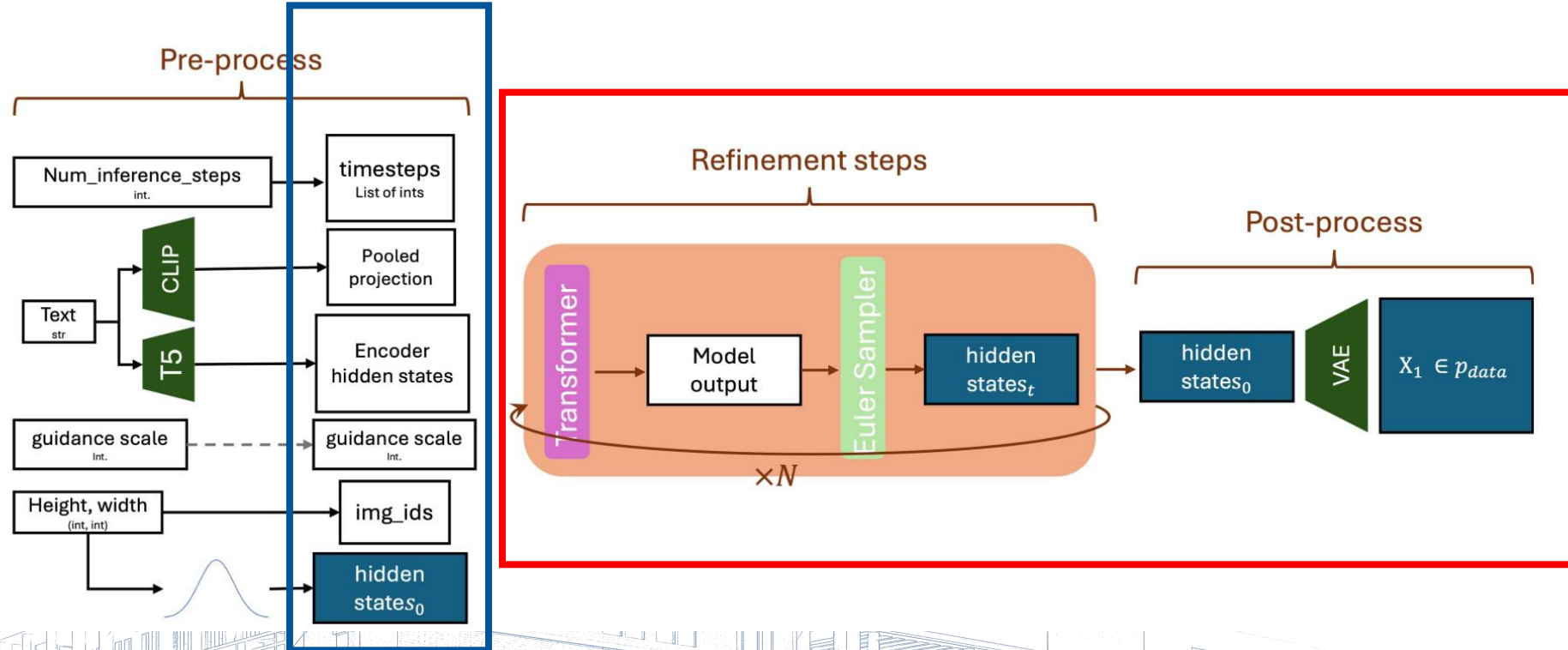
- **Dual Text Encoders**

- **CLIP:** prompts→class level or dense, high-dimensional embeddings
- **T5:**Transformer-based→rich, token-level semantic representations

- **Pre-process**

- text→pooled embeddings(CLIP)+dense embeddings(T5)
- num_inference_steps→ $t \in [0:T]$, typically $t=1000$
- resolution→spatial dimensions of latent noise sample $z_0 \sim N(0,1)$ + img_ids patch $(t, \hat{h}, \hat{w})$
- guidance_scale
- text_ids = $n \cdot (0, 0, 0)$, $n=512$ (T5)

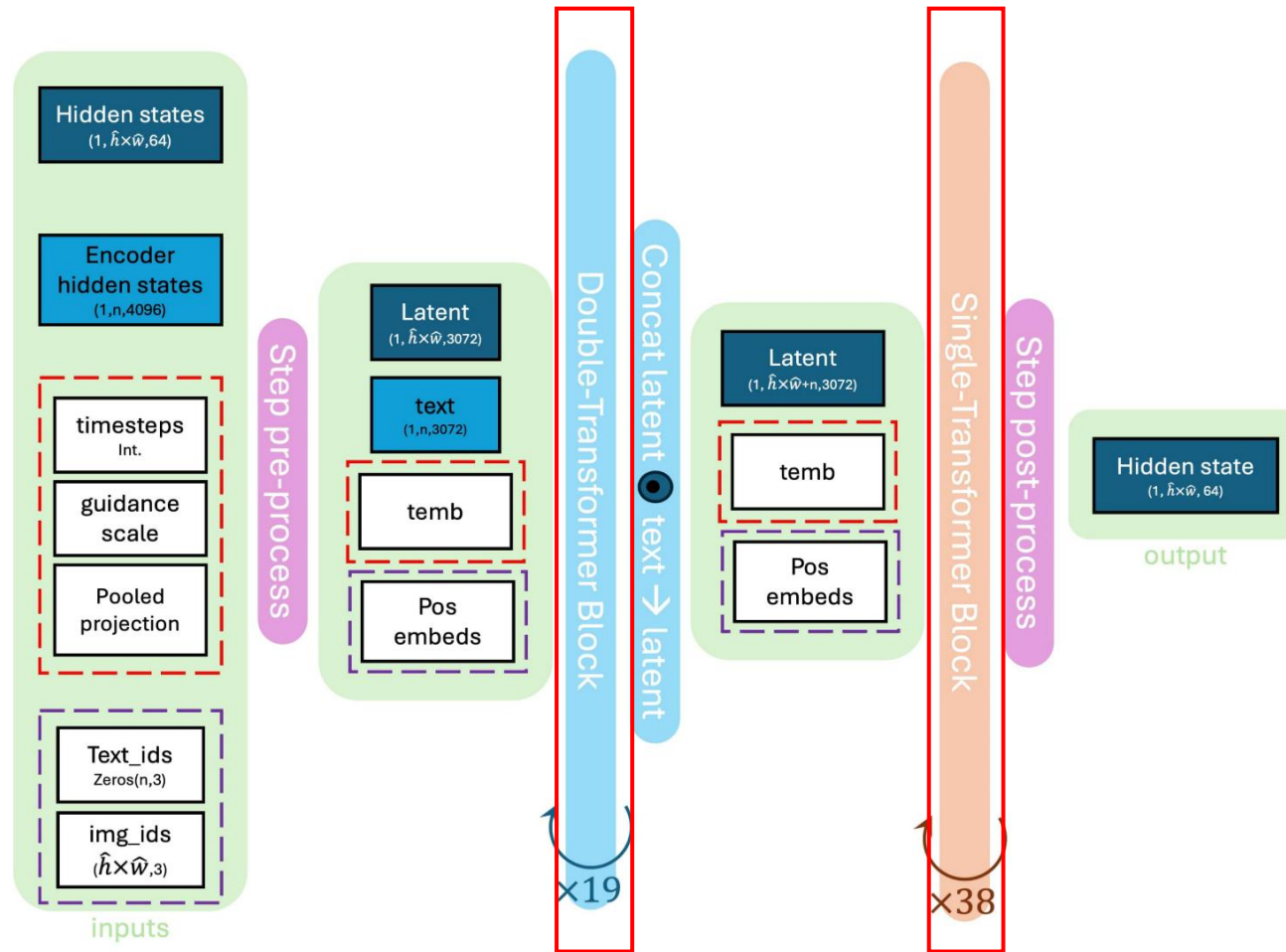
➤ Iterative Refinement and Post-processing



$$\Phi = [\text{encoder_hidden_states}, \text{pooled_projection}, \text{guidance}, \text{img_ids}, \text{text_ids}]$$

$$\forall t \in \text{timesteps} : z_{t+\Delta t} = \text{Samp}(v_{\theta}(z_t, t, \Phi))$$

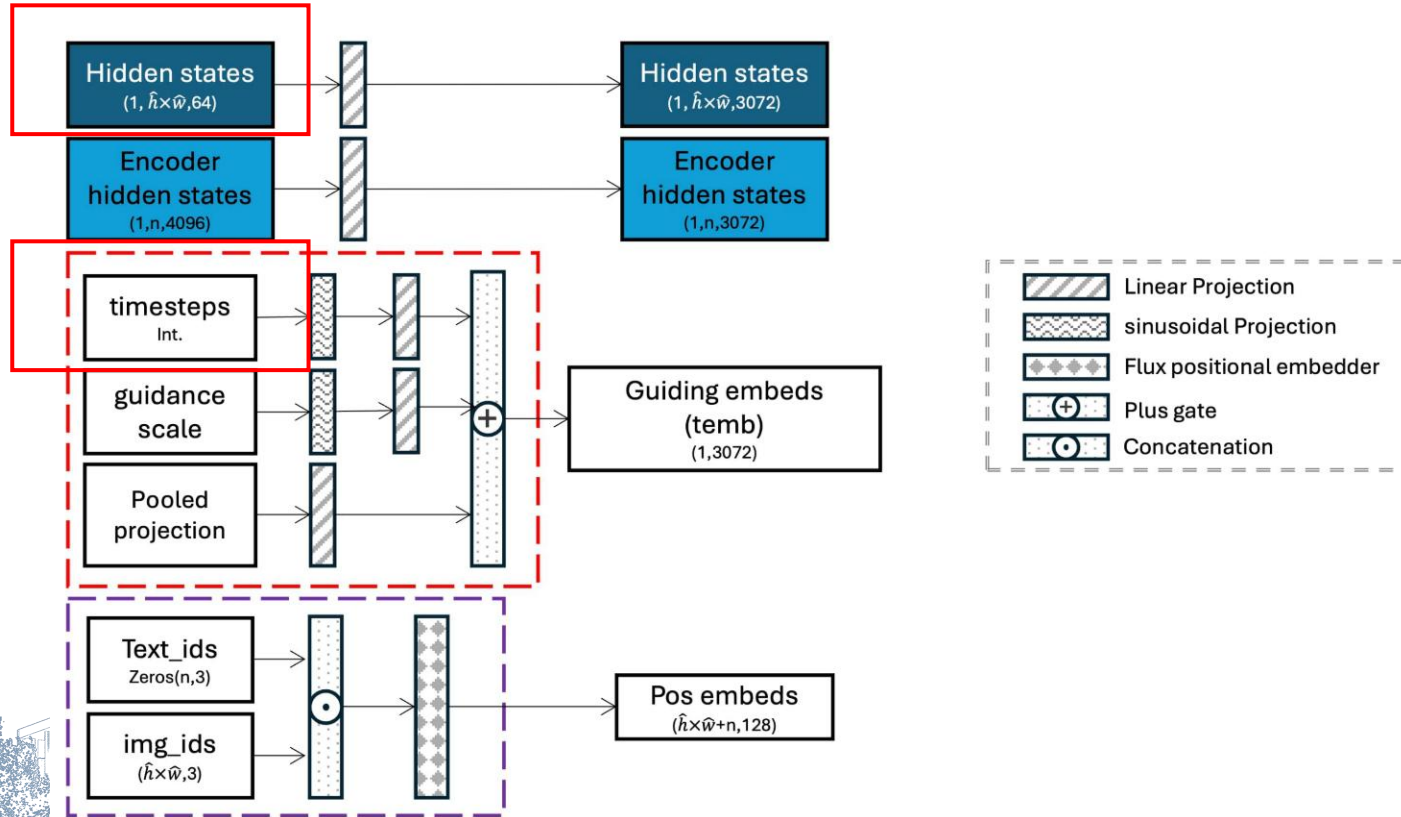
➤ Architecture of v_θ (Transformer Block)



- Per-iteration Pre-process
- Multi-Modal Attention
 - Double-Stream Transformer
 - Single-Stream Transformer

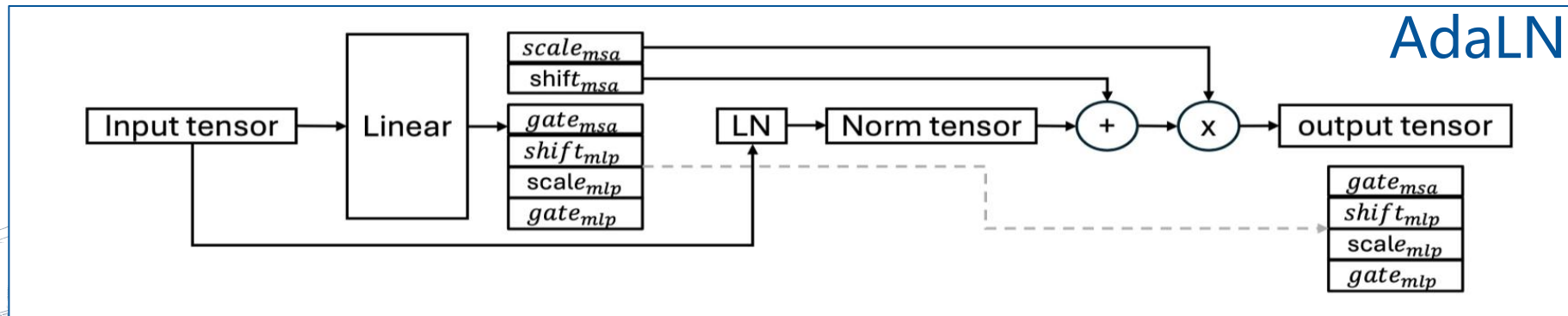
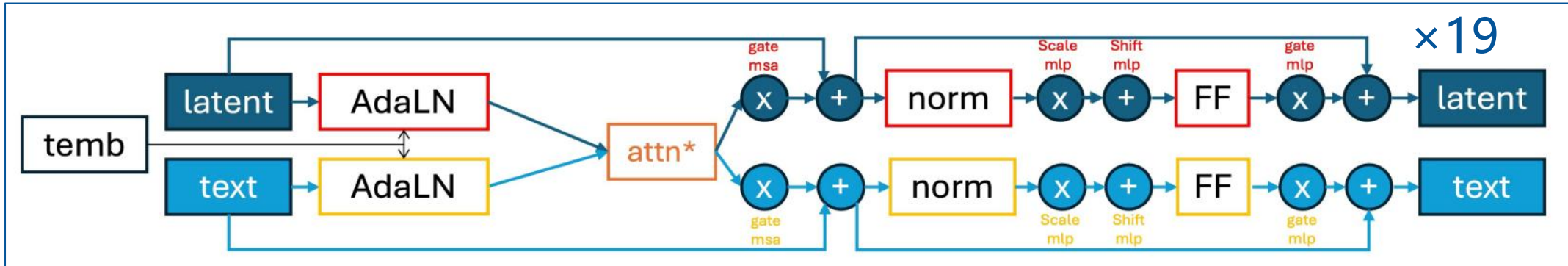


➤ Per-Iteration Pre-process



- Updated Parameters
 - timestep
 - Hidden states z_t
- Pre-process Steps
 - Shared dimensionality
 - Construction of guiding embeds
 - Positional embeds(constant)

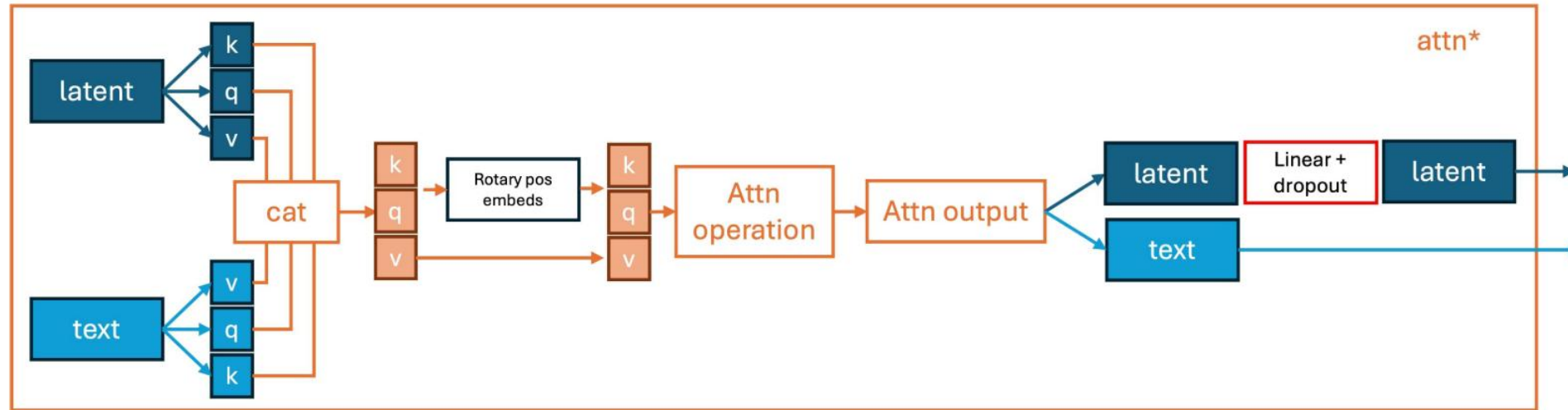
➤ Double-Stream Transformer Block



- AdaLN(Adaptive Layer Normalization) : Dynamically generates 6 Modulation Parameters
 - $scale_{msa}$, $shift_{msa}$, $gate_{msa}$
 - $scale_{mlp}$, $shift_{mlp}$, $gate_{mlp}$



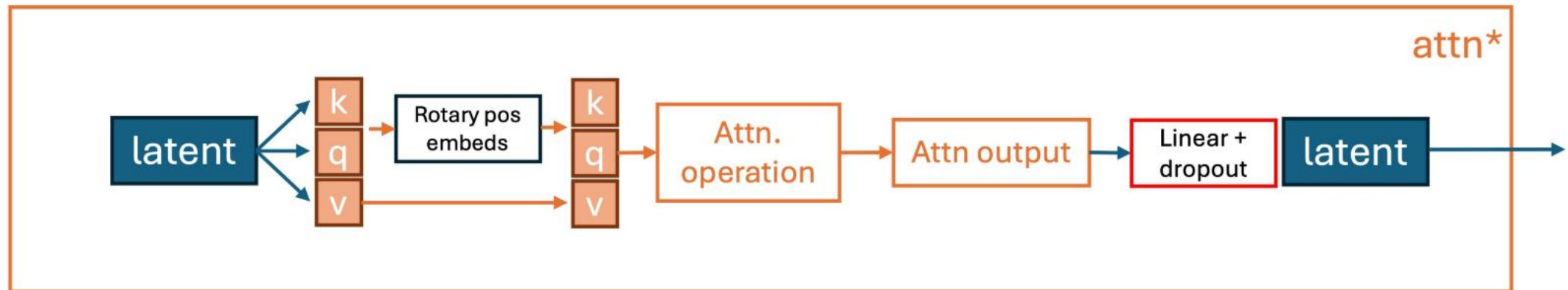
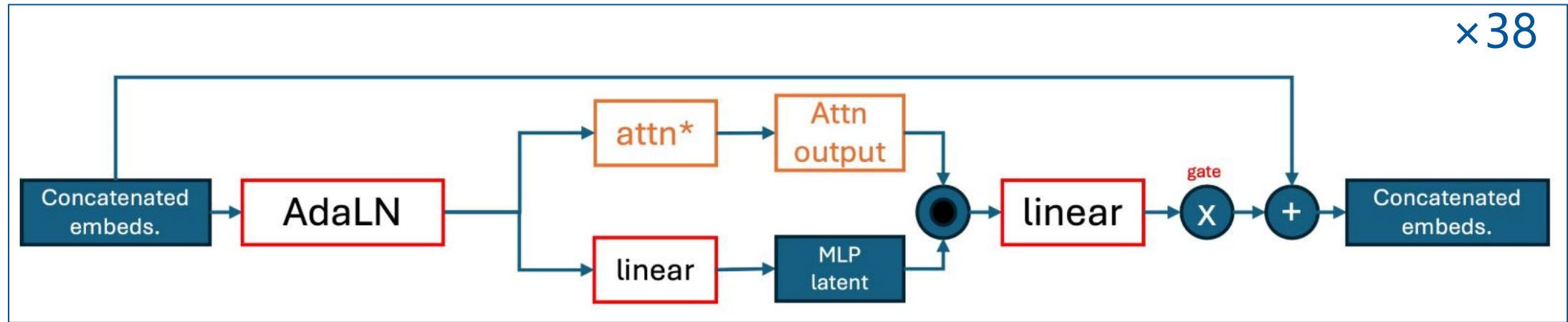
➤ Double-Stream Transformer Block



- **Multi-modality:** Applying attention to concatenated tokens.
- **RoPE:** Preserves relative positional relationships across sequences.
 - **Pre-calculate Rotation Parameters:** $(t, h, w) \rightarrow$ Sinusoidal frequencies $\rightarrow$ Cos and Sin tensors
 - **Rotate Q and K**
 - **Inject Relative Position:** Naturally fuses spatial information into the vector interaction.

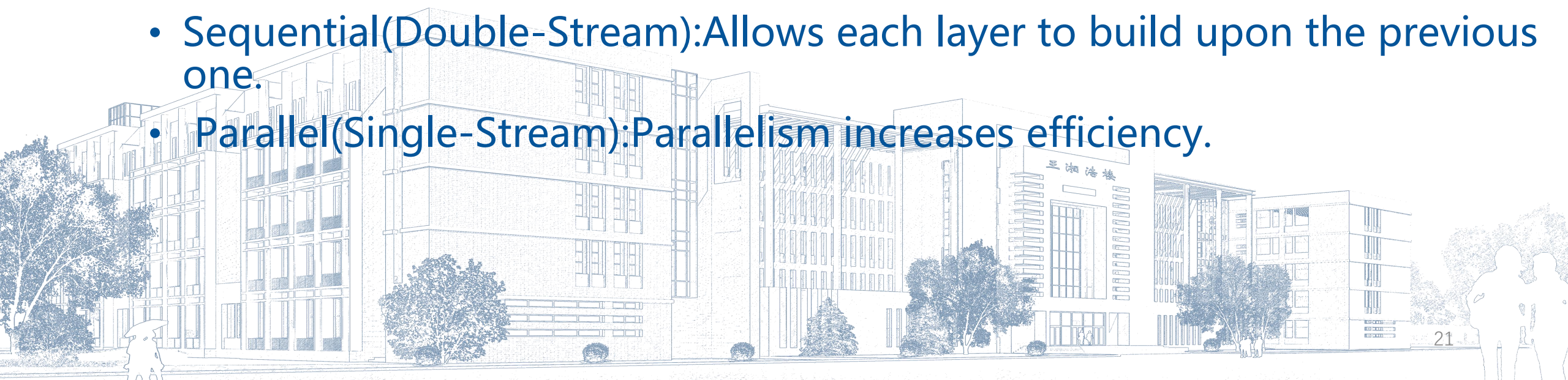


➤ Single-Stream Transformer Block



➤ Double-Stream vs Single-Stream

- Weight Sharing (Shared vs. Not Shared)
 - Shared(Single-Stream):Promote tighter integration between modalities.
 - Not Shared(Double-Stream):Specialize model representations for text and latent tokens independently.
- Computation Style (Sequential vs. Parallel)
 - Sequential(Double-Stream):Allows each layer to build upon the previous one.
 - Parallel(Single-Stream):Parallelism increases efficiency.



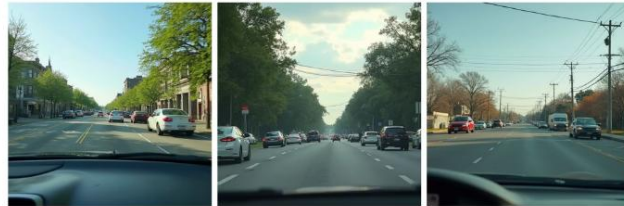
04

Qualitative Evaluation

FLUX.1 vs Stable Diffusion (SD)2.1

- Overall image quality

FLUX.1



Cars driving down the street on a **clear day**, taken from a vehicle's front dashcam. natural, realistic

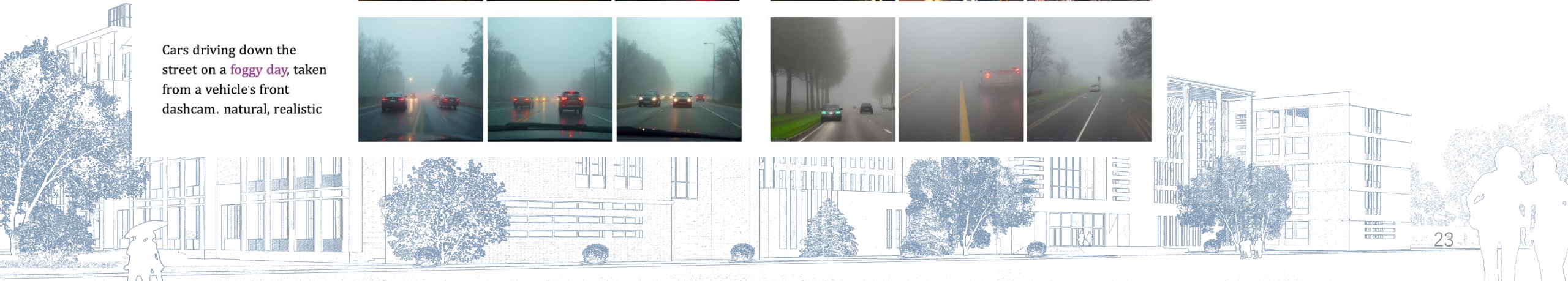
Stable Diffusion 2.1



Cars driving down the street on a **clear night**, taken from a vehicle's front dashcam. natural, realistic



Cars driving down the street on a **foggy day**, taken from a vehicle's front dashcam. natural, realistic



FLUX.1 vs Stable Diffusion (SD)2.1



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- Text-image alignment

“... a slightly snowy day...”

FLUX.1



“...a heavily snowy day...”



“...sunny day after a snowy period...”



Stable Diffusion 2.1



FLUX.1 vs Stable Diffusion (SD)2.1

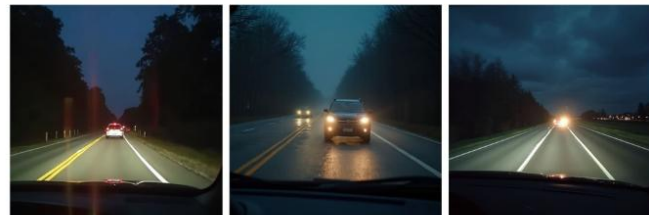
- Alignment to sparse prompts: text descriptions that are minimal or loosely defined.

FLUX.1



A car is driving down the road in an adverse weather, in extremely difficult viewing conditions. taken from a vehicle's front dashcam

Stable Diffusion 2.1



A car is driving down the road in low light with difficult viewing conditions. taken from a vehicle's front dashcam



A car is driving down the road in low light, **blurry**, difficult viewing conditions. taken from a vehicle's front dashcam



FLUX.1 vs Stable Diffusion (SD)2.1



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- Synthesis of rare concepts

“...an upside-down crashed car...”

FLUX.1



“...a boat trailer...”



Stable Diffusion 2.1



FLUX.1 vs Stable Diffusion (SD)2.1



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- Adherence to complex prompts containing multiple descriptive elements.

FLUX.1

Stable Diffusion 2.1

“Cars driving down a crossroad on an **overcast day**,
an **old man on a wheelcahir** and a **toddler** are
crossing the road.”



“Cars driving down a street **crowded with people**
on a clear day, **taken from a vehicle's front
dashcam.**”



“An automotive scene **taken from a car's
front dashcam**. A **red motorcycle** is speeding
up the road, chased by a **police car.**”



05

Conclusion

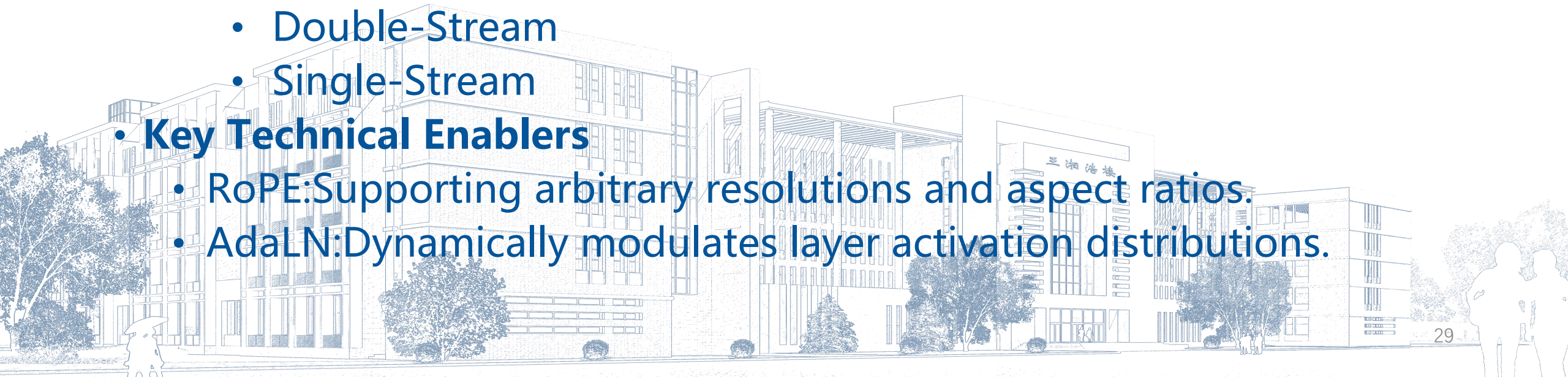


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Conclusion

- **Paradigm Shift:**Traditional denoising diffusion→Rectified Flow
- **Hybrid Transformer Architecture**
 - U-Net→Full Transformer
 - Dual Text Encoders (CLIP+T5)
 - Unique Block Composition
 - Double-Stream
 - Single-Stream
- **Key Technical Enablers**
 - RoPE:Supporting arbitrary resolutions and aspect ratios.
 - AdaLN:Dynamically modulates layer activation distributions.





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THANKS

